Enhancing Visitor Experiences at Lapworth Museum

A Comprehensive Data Analytics Approach to User Feedback

Saloni Kamble
MSc Business Analytics
2024

Client: Lapworth Museum of Geology

UNIVERSITY OF BIRMINGHAM | BIRMINGHAM BUSINESS SCHOOL

Executive Summary

This report analyses visitor feedback for the Lapworth Museum to identify sentiment patterns and suggest improvements. Using data from Google Reviews, TripAdvisor, and the museum's own reviews, we performed a sentiment analysis, revealing that while overall visitor sentiment is highly positive, there are notable areas for improvement.

To provide context, we benchmarked the Lapworth Museum against the Manchester Museum and the Barber Institute of Fine Arts, which highlighted areas where the Lapworth Museum could enhance its visitor experience.

Our recommendations focus on addressing these pain points. We suggest improving parking facilities, installing vending machines for visitor convenience, and enhancing signage for better navigation. Additionally, we propose implementing a feedback loop using Net Promoter Scores (NPS) to regularly assess visitor satisfaction and guide continuous improvements.

By acting on these recommendations, the Lapworth Museum can further elevate the visitor experience, maintaining its reputation as a leading cultural destination.

Contents

Executive Summary 2
Acknowledgements 4
1. Introduction 5
1.1 Objectives of the Project 5
1.2 Summary of Results 5
2. Literature Review 7
2.1 Sentiment Analysis on User Feedback 7
2.2 Benchmarking Analysis 7
2.3 Application of Visualization 7
2.4 Feedback Loop 8
3. Methodology 9
3.1 Research Design Justification 9
3.2 Methods and their Applications 9
3.3 Data Collection and Preprocessing 9
3.4 Strengths and Limitations 10
3.5 Ethical Considerations 10
4. Analysis and Findings 11
4.1 Sentiment Analysis of Visitor Feedback 11
4.1.1 Lapworth feedback data 11
4.1.2 Google Maps and TripAdvisor 13
4.2 Benchmarking 17
4.2.1 Overview of the sentiment analysis 18
4.2.2 Comparative Analysis: Lapworth Museum vs Manchester Museum 18
4.2.3 Comparative Analysis: Lapworth Museum vs. Barber Institute of Fine Arts 19
4.3 Visualization 20
5. Recommendations 26
5.1 Short-term Solutions 26
5.2 Long-Term Plan: 27
5.2.1 Feedback loop 28
5.2.2 Building Community and Engagement within Visitors 29
6. Conclusion 30
7. References 31

Acknowledgements

We would like to extend our sincere gratitude to Jon Clatworthy, the Director at Lapworth Museum, whose collaboration and insights were invaluable to the success of this project. His commitment to the museum and its mission greatly inspired our work.

We are also deeply appreciative of Angela George, the data analyst at Lapworth, for providing us with all the necessary data and for her prompt and thorough responses to our queries. Her expertise and support were crucial to the completion of our analysis.

Our thanks also go to Anupam Mehta, whose guidance and constructive feedback were instrumental throughout the research process. We greatly value the time and effort invested in helping us refine our approach.

Finally, we would like to thank our colleagues and peers for their helpful suggestions and encouragement, which contributed significantly to the development of this project.

1. Introduction

Museums are heralded as community treasures, bringing people together to explore and engage with shared history and culture. Among these, the Lapworth Museum, often celebrated as Birmingham's best "hidden gem" (Brinkworth, 2024), has gained recognition for its dinosaurs, fossils, and more.

Despite the challenging landscape of recent years, Lapworth Museum has demonstrated remarkable business resilience. Visitor numbers have steadily increased, a testament to the museum's adaptability and enduring appeal, even after being closed during the pandemic.

From our sentiment analysis, we can see that Lapworth has reputational resilience as well. The museum has garnered an overwhelming number of positive reviews across platforms like TripAdvisor and Google Reviews, underscoring its continued success.

This report presents the findings of an in-depth data analytics capstone project conducted for Lapworth Museum. By analysing user feedback, this project uncovers key trends, patterns, and actionable insights that will inform strategic management decisions, ultimately enhancing the overall visitor experience.

1.1 Objectives of the Project

The project aimed to achieve the following objectives:

- **Conduct a comprehensive sentiment analysis** to categorize user feedback as positive, negative, or neutral, and provide an overall sentiment score for the museum.
- **Identify key themes and trends** within the user feedback to understand the main factors influencing visitor experiences.
- **Develop targeted recommendations** for management to enhance positive visitor experiences and address any concerns highlighted in the feedback.
- **Benchmark Lapworth Museum's user feedback** against industry standards and similar institutions to identify areas of competitive advantage or opportunities for improvement.
- **Propose a sustainable feedback loop strategy** that encourages continuous user engagement and supports ongoing improvements based on visitor preferences.
- **Utilize data visualization tools** to present findings clearly, enabling effective decision-making for museum management.

1.2 Summary of Results

- **Sentiment Analysis:**

  The analysis of user feedback reveals that Lapworth Museum enjoys a strong positive reputation, with 75.15% of feedback being positive and an **overall sentiment score of 0.368**.

  Reviews from Google and TripAdvisor are even more favourable, with 85.76% positive feedback and a **sentiment score of 0.339**.

  While the majority of visitors had a positive experience, there are opportunities to enhance the museum's offerings, particularly in areas with neutral feedback (23.04% overall) and a small amount of negative feedback (1.81% overall).

- **Benchmarking:**

  For benchmarking, we selected Manchester Museum and the Barber Institute of Fine Arts to provide a comparative perspective on sentiment scores.

  Manchester Museum scores **0.28**, with recent renovations acknowledged; however, some visitors felt these changes were inadequate and certain exhibits lacked content.

  The Barber Institute of Fine Arts, with a slightly lower score of **0.26**, is praised for its warm atmosphere and knowledgeable staff, though it receives feedback about the limited size of its collection.

  Lapworth Museum score of **0.28** is on par with Manchester Museum and slightly higher than Barber, showing that despite limited funds, Lapworth is performing relatively better.

- **Recommendations:**

  Recommendations are divided into two parts: short-term and long-term solutions. Short-term solution are quick fixes that the management can action within weeks. It should take few months to implement. These solutions include improving signage, providing adequate parking near the museum, and installing a vending machine.

  Long-term solutions focus on ensuring a sustainable future for the museum. These include implementing feedback loops and developing strategies to build and engage the community.

2. Literature Review

2.1 Sentiment Analysis on User Feedback

Sentiment analysis is a widely used natural language processing technique designed to identify and categorize opinions expressed in text. It is particularly valuable for understanding customer feedback, as it enables organizations to quantify subjective data and uncover underlying sentiments.

According to (Liu, 2020), sentiment analysis can be applied at various levels, including document, sentence, and aspect levels, each offering different insights into user attitudes and emotions.

Incorporating online reviews such as Google and TripAdvisor into sentiment analysis can help organizations in decision-making, analysing customer behaviour, and collecting insightful information that cannot be gathered manually (Jain, et al., 2021).

Techniques for sentiment analysis typically fall into two broad categories: lexicon-based approaches and machine learning approaches. Lexicon-based methods rely on predefined lists of words associated with positive or negative sentiments, while machine learning techniques involve training models on labelled data to predict sentiment.

We chose a lexicon-based approach because user feedback is primarily text-based, and this method is well-suited for analysing the sentiment expressed in written reviews.

2.2 Benchmarking Analysis

Benchmarking is a strategic management tool used by organizations to measure their performance against industry standards or competitors (Stevenson, 1996).

In the context of cultural institutions, benchmarking allows museums to compare their visitor satisfaction, operational efficiency, and service quality with those of similar organizations. This comparison can identify gaps in performance and highlight opportunities for improvement (Freytag & Hollensen, 2001).

In museums, benchmarking can lead to enhanced visitor experiences, more efficient operations, and a better understanding of how to engage audiences effectively. This process is particularly valuable for museums aiming to maintain competitiveness in a crowded cultural landscape by continuously refining their strategies and offerings.

2.3 Application of Visualization

Visualization plays a critical role in the analysis and presentation of visitor feedback, enabling organizations to quickly identify and communicate key insights (Unwin, 2020).

Word clouds, for example, are a popular tool for thematic exploration as they visually represent the most frequently occurring terms in a body of text. McNaught (2010) highlights that word clouds can be an effective starting point for identifying key themes in visitor feedback.

Additionally, visualizations are essential for summarizing data in a way that is accessible to a broad audience, including museum management and stakeholders. These tools allow for the quick identification of trends, patterns, and outliers in visitor feedback, facilitating data-driven decision-making.

2.4 Feedback Loop

The literature further suggests that implementing feedback loops, as proposed in our recommendations, can significantly enhance visitor engagement and satisfaction. In recent years, the concept of feedback loops has gained significant attention in service-oriented industries as a means to foster continuous improvement. A well-designed feedback loop is essential for adapting to changing circumstances and ensuring that interventions are both effective and responsive to user needs.

According to Viswanath et al. (2019), an effective feedback loop should exhibit agility, adaptability, and an iterative process, allowing for multiple rounds of feedback and refinement. It should also be responsive to the needs of stakeholders and actionable, providing timely and relevant data to inform decision-making.

In the context of museums, implementing such feedback loops can enhance visitor engagement and satisfaction by continuously refining services based on direct input from users. This approach ensures that feedback is not only collected but also meaningfully integrated into the museum's operations, leading to sustained improvements and a more personalized visitor experience.

3. Methodology

3.1 Research Design Justification

The research design focused on gaining a comprehensive understanding of visitor feedback at the Lapworth Museum. An exploratory data analysis (EDA) combined with sentiment analysis was chosen for its ability to uncover patterns and sentiments in qualitative data, informing strategic decisions.

This mixed-method approach offers a deeper understanding of visitor experiences. Sandelowski (2000) strongly supports the use of mixed-method approaches, highlighting that the integration of qualitative and quantitative methods can enhance the depth, validity, and applicability of research findings. Sentiment classification and theme identification provide context, while sentiment scoring and benchmarking offer measurable performance indicators for the museum.

3.2 Methods and their Applications

- **Sentiment Analysis:** TextBlob the Python library was employed, which utilizes a rule-based method for sentiment classification.
  It assigns a polarity score to a piece of text, which ranges from -1 to 1. **A score of 0 represents neutral sentiment, while 1 indicates positive sentiment, and -1 signifies negative sentiment.** TextBlob evaluates each word in the text and assigns a semantic score, which is then combined to produce a weighted average representing the overall sentiment of the sentence based on word polarities.
  In addition to polarity, TextBlob also calculates subjectivity, a value between 0 and 1. A score of 0 indicates the text is objective, relying on factual information, while a score closer to 1 suggests the text is subjective, reflecting personal feelings, emotions, or opinions (Zahoor & Rohilla, 2020). TextBlob's capability to provide a polarity score for each piece of feedback allowed us to quantify sentiment.
- **Thematic Analysis:** To identify recurring themes and trends in the feedback, we utilized word clouds. Word clouds were generated to visualize the most frequently occurring terms in user reviews, aiding in the identification of key aspects influencing visitor experiences, as they help highlight prominent themes and recurring topics (Heimerl, et al., 2014).
- **Comparative Benchmarking:** To place the Lapworth Museum's feedback in context, we compared sentiment scores with industry benchmarks and feedback from similar institutions. This approach is grounded in comparative analysis methodologies commonly used in benchmarking studies, where the performance of an entity is assessed against industry standards or competitors.
- **Data Visualization:** The insights derived from the analysis were presented through tableau visualization tool, including scatter plots and word clouds. This approach is supported by literature that emphasizes the importance of visual analytics in enhancing comprehension and facilitating data-driven decision-making.

3.3 Data Collection and Preprocessing

The data was collected from three sources: Lapworth Museum's internal survey data and user reviews from Google Maps and TripAdvisor. The museum's survey data, originally provided in separate files for each year, was consolidated after excluding outdated or incomplete entries, resulting in a total of approximately 1,800 feedback entries. This included both three-word descriptions of the experience and more detailed written feedback. External data was extracted using a web scraping tool, allowing

us to gather 640 user reviews from Google Maps and TripAdvisor, platforms where users often express their genuine opinions.

To ensure high-quality results, we conducted a thorough sentiment cleaning process. This involved removing unnecessary characters and stop words (e.g., ‘and’, ‘this’, ‘of’, and ‘it’). According to (Ghag & Shah, 2015), removing stop words increases the accuracy of polarity-based sentiment classifiers, improving performance from 50% to 58.6%. In addition, we expanded the list of stop words to include terms specific to our case, such as ‘museum’ and ‘university,’ which are common in the reviews but not useful for sentiment analysis.

Another key preprocessing step was lowercasing all text, as recommended by Zhang, et al., (2015), to ensure uniformity in the data. Correct spelling is also essential, as it helps the sentiment classifier perform more accurately, given that user feedback is written by humans and may contain typos.

Lemmatization, a technique used to reduce words to their base or root form, was also applied. This step ensures that different forms of the same word (e.g., 'amazing,' 'amaze,' 'amazes') are analysed as a single entity ('amaze'), which helps improve the accuracy of the sentiment analysis (Guzman & Maalej, 2014).

3.4 Strengths and Limitations

The research design's strength lies in its comprehensive approach, combining sentiment analysis from different sources, thematic exploration, and comparative benchmarking. This allowed for a robust analysis of visitor feedback, offering both qualitative and quantitative insights. However, the methodology also has limitations. The exclusion of the 2019-2020 data due to its incompleteness may have led to the loss of potentially valuable historical insights.

Additionally, while reliance on TextBlob for sentiment analysis is effective due to its simplicity and ease of use, it has limitations in capturing the nuanced meaning of the text, as it primarily uses a rule-based approach with less contextual understanding. More advanced models like VADER, could potentially offer more accurate sentiment classification by considering context and the complexities of language. However, VADER is optimized for social media text rather than longer feedback reviews as in Lapworth’s case (Prasad, et al., 2024). Given these constraints, TextBlob was chosen for its balance of accuracy and simplicity.

3.5 Ethical Considerations

Ethical considerations were paramount in the design and execution of this project. The data used for analysis was anonymized to protect the privacy of the individuals providing feedback. Additionally, the analysis was conducted with the intent to improve the visitor experience, aligning with ethical standards of beneficence and respect for persons. Furthermore, the use of publicly available data from Google Maps and TripAdvisor was done in accordance with the terms of service of these platforms, ensuring that the research adhered to legal and ethical guidelines.

4. Analysis and Findings

In this section, we will analyse the collected data and present the key findings related to visitor feedback for the Lapworth Museum. The analysis is structured according to the project objectives. The findings are supported by relevant data visualizations.

4.1 Sentiment Analysis of Visitor Feedback

We analysed data from Lapworth Museum, and to enrich our findings, we also examined reviews from Google and TripAdvisor. This approach provided additional context and nuanced insights, enhancing our overall analysis.

4.1.1 Lapworth feedback data

Figure 1 presents the sentiment distribution derived from the visitor feedback at the Lapworth data. The majority of feedback, 75.15%, is positive, indicating a generally favourable response to the museum's offerings.

Approximately 23.04% of the feedback is neutral, suggesting that while visitors may not have had a strong emotional reaction to their experience, it was neither negative nor particularly impactful.

Only 1.81% of the feedback is classified as negative, which reflects a small minority of visitors who had less satisfactory experiences. **The overall sentiment score is 0.368.**

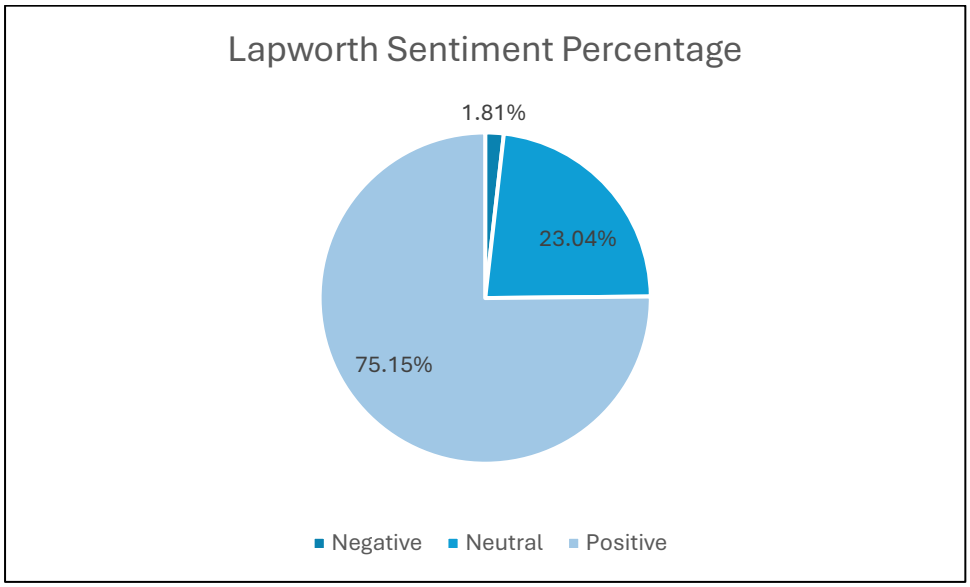

Figure 1 Lapworth Sentiment Distribution

Figure 2 presents a word cloud generated from the positive feedback received by the Lapworth Museum. The most frequently mentioned terms provide clear insights into the aspects of the museum that visitors appreciate the most. The dominant word in the word cloud is "educational" suggesting that visitors highly value the museum’s educational offerings. This is further supported by related terms such as "informative" and "knowledgeable" indicating that visitors see the museum as a place of learning and discovery. Other prominent words include "fun" "engaging" and "entertaining" emphasizing that visitors find the museum experience enjoyable and interactive. This balance between education and entertainment is a key strength of the museum and a crucial factor in its positive reception.

Additionally, terms such as "wonderful" "fantastic" and "amazing" suggest that many visitors had exceptional experiences that exceeded their expectations. These enthusiastic terms reinforce the museum's reputation as a place of high-quality exhibits and enjoyable visits.

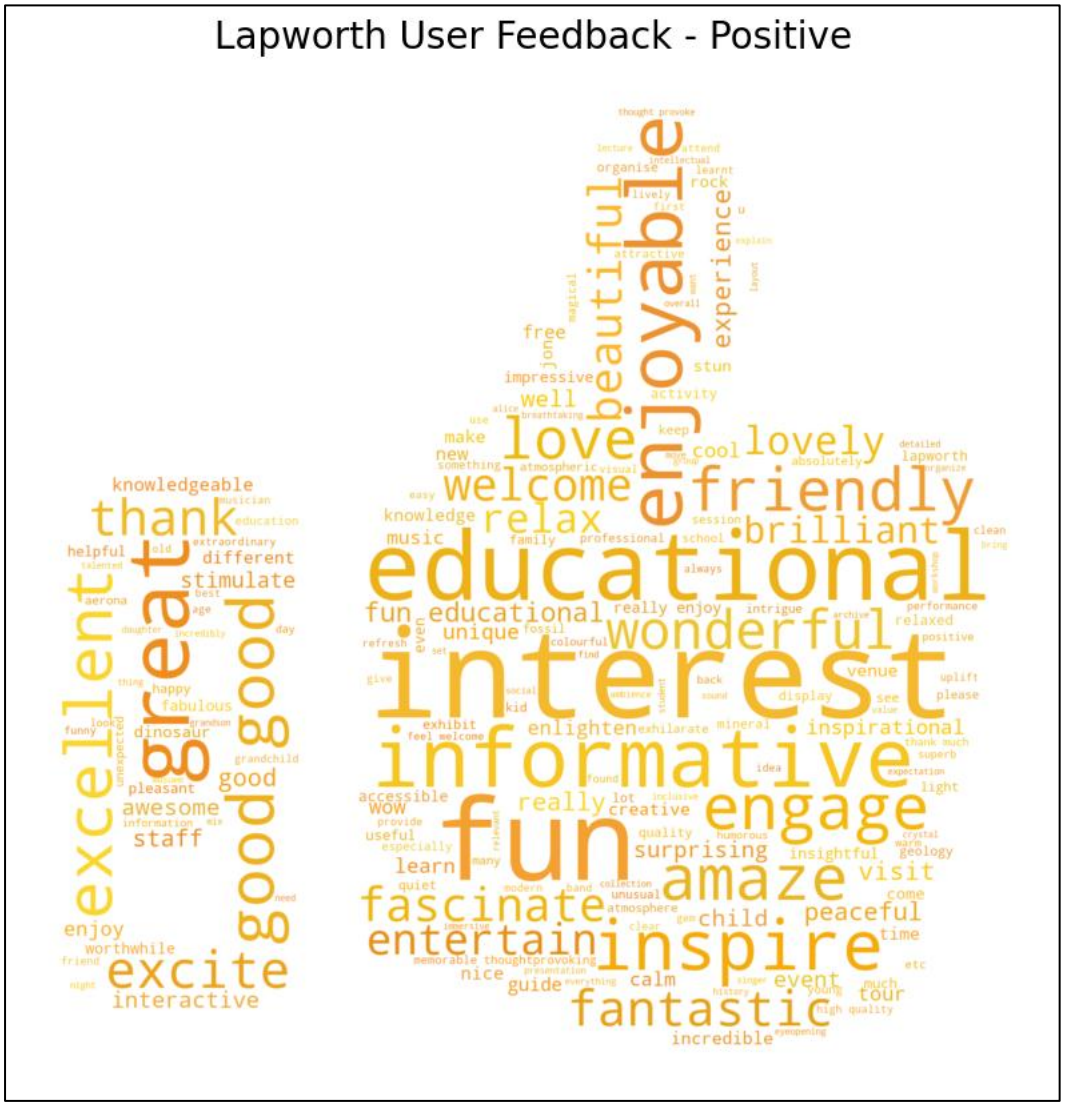

Figure 2 Positive Lapworth User Feedback word cloud

Figure 3 displays a word cloud generated from the negative feedback provided by visitors to the Lapworth Museum.

The most significant term is "find" closely associated with words like "difficult" and "signpost" This indicates that visitors consistently experienced challenges in locating the museum, likely due to insufficient signage or unclear directions.

Another commonly mentioned term "cold" suggests that some visitors found the museum environment to be uncomfortable or not sufficiently inviting. This feedback highlights the need to improve environmental factors such as temperature control or ambiance to create a more pleasant atmosphere for visitors.

Words like "boring" and "poor" indicate that certain exhibits or aspects of the museum failed to engage some visitors.

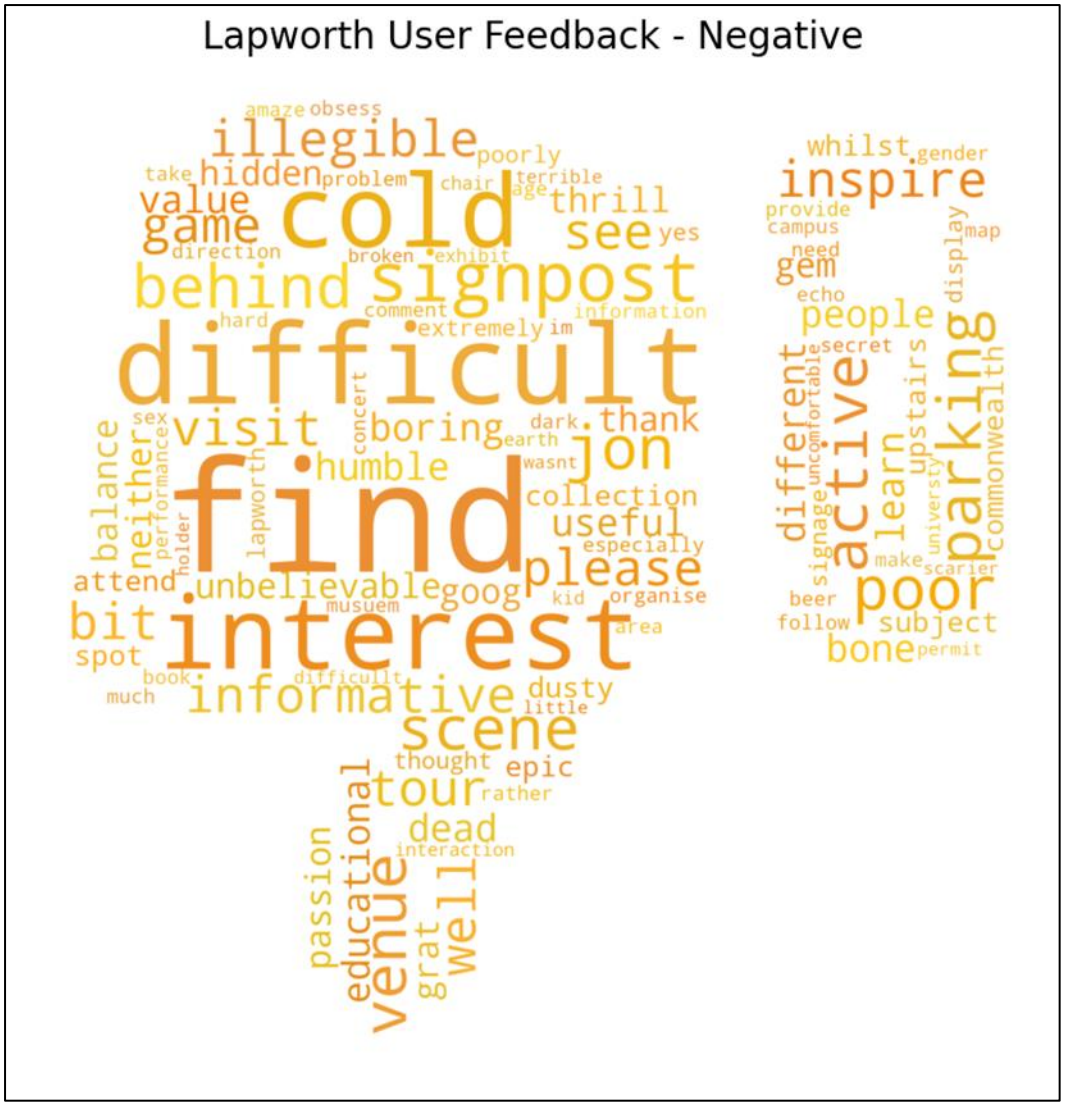

Figure 3 Negative Lapworth User Feedback word cloud

4.1.2 Google Maps and TripAdvisor

The pie chart in Figure 4 illustrates the sentiment distribution of user reviews for the Lapworth Museum collected from Google and TripAdvisor.

The overwhelming majority of the feedback, 85.76%, is positive. This indicates that most users who reviewed the museum on these platforms had a favourable experience. This strong positive sentiment highlights the museum's success in delivering an enjoyable and valuable visitor experience.

Approximately 11.89% of the feedback is neutral. This suggests that a portion of visitors had experiences that neither significantly impressed nor disappointed them. It is an opportunity for the museum to consider how it can convert neutral experiences into more memorable and positive ones by enhancing areas that may currently be underwhelming.

Only 2.35% of the feedback is negative. While this is a small fraction, addressing these issues, could help further reduce the percentage of negative feedback. **The overall sentiment score of Google and TripAdvisor reviews is 0.339.**

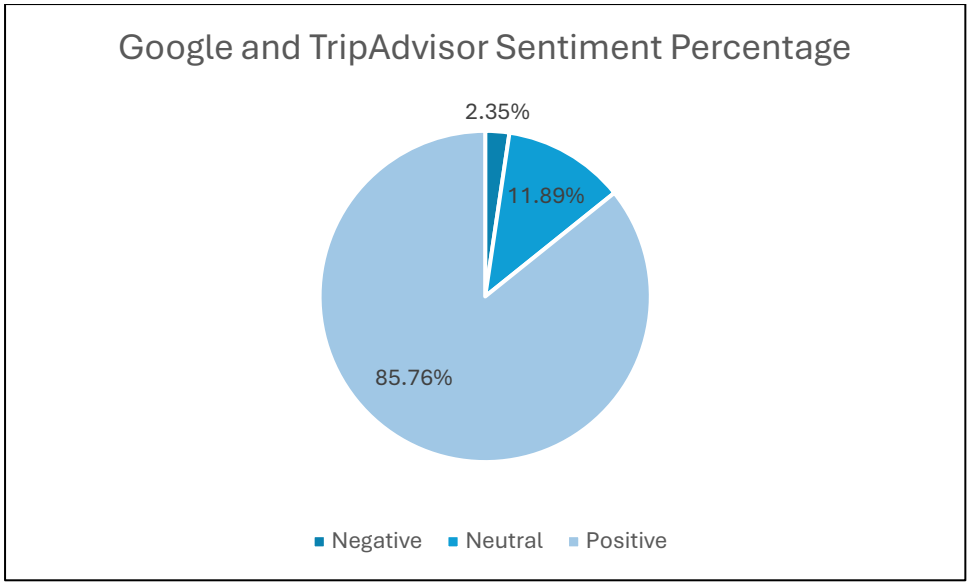

Figure 4 Google and TripAdvisor Sentiment Distribution

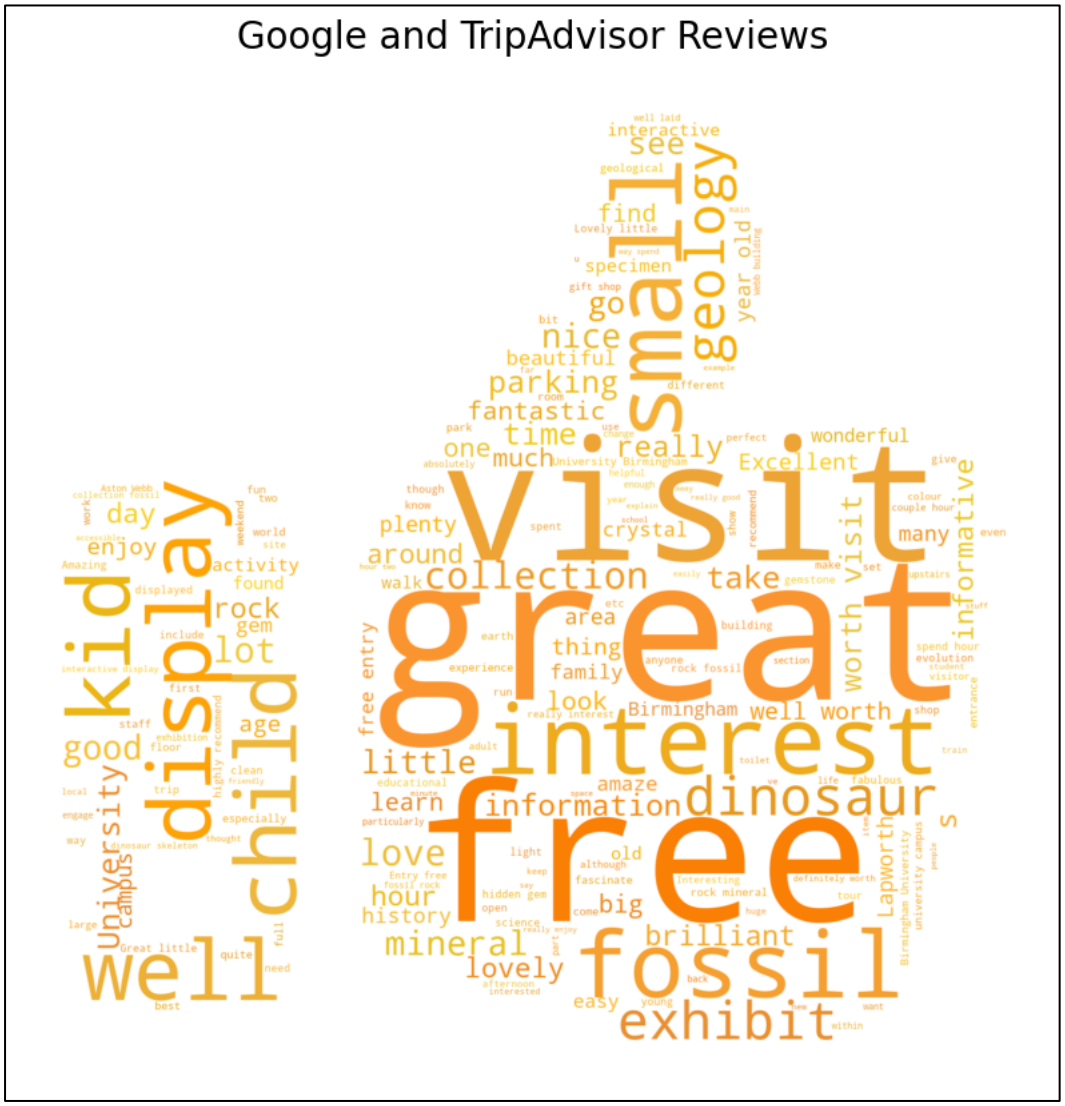

Figure 5 Positive Google and TripAdvisor Word cloud

Figure 5 shows the positive word cloud with the most frequently occurring terms from Google and TripAdvisor reviews about the Lapworth Museum.

The most prominent word is “great” indicating that many visitors describe their overall experience at the museum in positive terms. Another key term is “free” which highlights that visitors highly value free entry to the museum. This likely contributes significantly to the positive sentiment, as cost is often a deciding factor in visitor satisfaction.

The frequent appearance of words like “fossil” “geology” and “mineral” reflects visitors' appreciation for the museum's natural history and geological exhibits.

The words “kid” “child” and “dinosaur” indicate that the museum is perceived as a family-friendly venue. This emphasis on kid-friendly content further strengthens the museum’s appeal to families.

Terms like “visit” “informative” and “collection” further underscore the value visitors find in the museum’s educational exhibits and their overall experience.

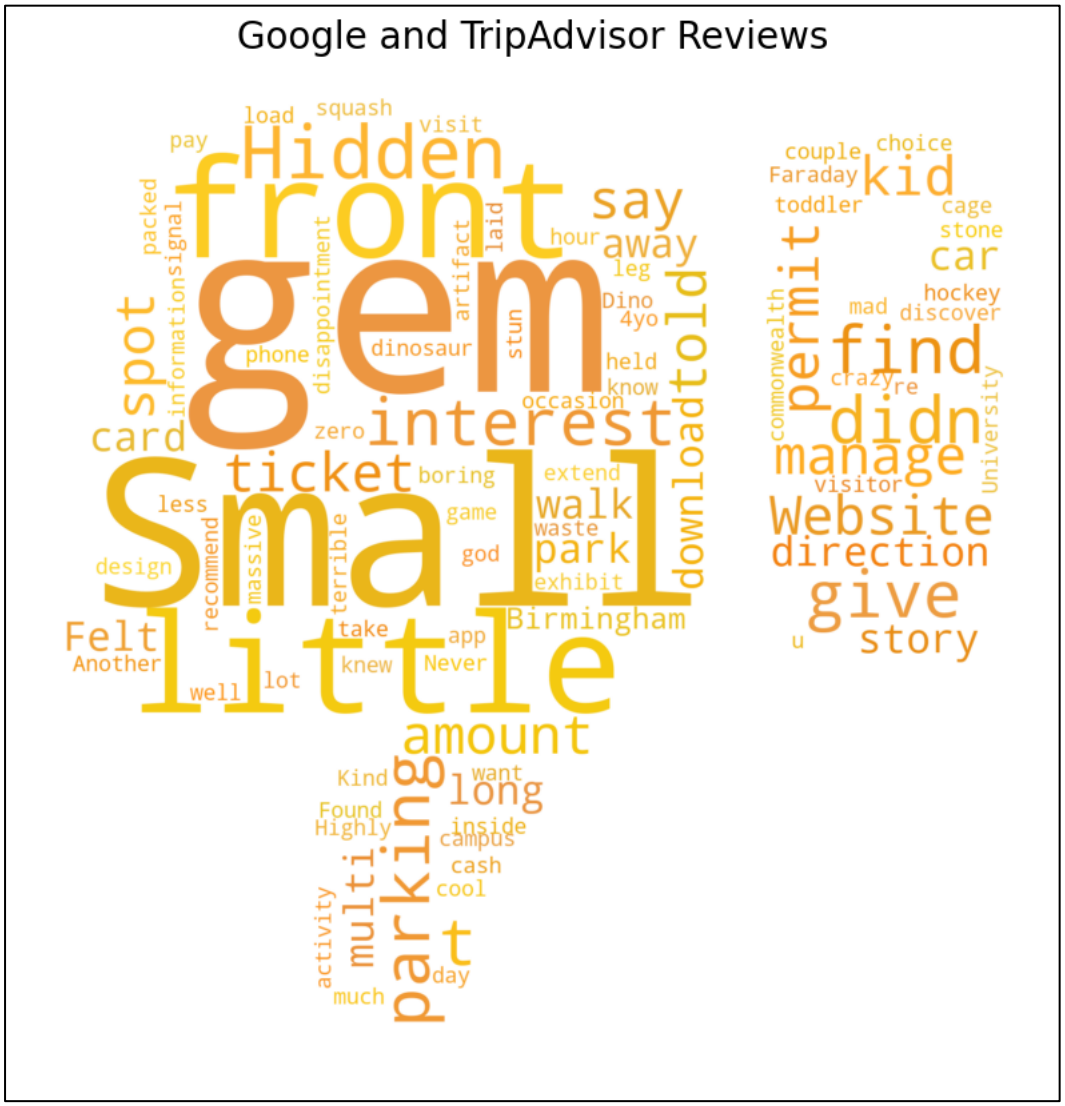

Figure 6 Negative Google and TripAdvisor Word cloud

Figure 6 shows the word cloud captures the recurring negative themes from Google and TripAdvisor reviews about the Lapworth Museum. Several key issues stand out prominently and provide insight into visitor dissatisfaction.

The term "gem" appears frequently but is likely used in combination with other words such as "hidden" and "small" suggesting that while some visitors may appreciate the museum as a hidden gem, others might find it too obscure or difficult to locate. The words "small" and "little" reflect concerns about the museum’s size and visibility, implying that some visitors found the exhibits limited or the museum itself hard to find or underwhelming in scale.

The words "parking" and "permit" also appear prominently, highlighting frustrations related to parking facilities or regulations. Visitors might have had trouble finding adequate parking or navigating the parking system.

Other notable terms like "boring" and "disappointment" point to potential dissatisfaction with either the exhibits or the overall experience. These terms suggest that a segment of visitors found the museum's offerings less engaging than expected.

4.2 Benchmarking

“Benchmarking is a technique used to compare key figures and rank organizations against competitors or industry averages. It involves measuring performance levels, comparing procedures, learning from benchmarking partners, and ultimately aiming for improvement in the business through benchmarking studies.” (Bala Krishnamoorthy, January 2014)

In this analysis, we compare Lapworth Museum's user feedback with two key benchmarks: the Manchester Museum and the Barber Institute of Fine Arts.

The Manchester Museum was selected as a benchmark due to its scale and diverse range of exhibits, including natural history, anthropology, and archaeology, providing a comparison of how a multidisciplinary museum engages and retains visitors. Another point of commonality is that both the Lapworth and Manchester Museum are part of prestigious universities. This shared academic affiliation makes Manchester Museum is a great example to evaluate how two university-owned museums use academic resources, research, and educational outreach into their exhibitions and visitor experiences.

Conversely, while being smaller, **The Barber Institute of Fine Arts** was picked due to its specific focus on the arts, much to Lapworth's geology specialism. The Barber Institute is renowned for its art collection, attracting a dedicated visitor base, this makes it an excellent comparison for understanding how a specialized museum can create a unique and engaging visitor experience. Additionally, both the places are supported by the university of Birmingham, therefore being funded by the same source.

A total of 180 reviews, combining feedback from both TripAdvisor and Google Reviews were used for the Lapworth, Manchester Museum, and the Barber Institute. This mix of sources provided a comprehensive view of the sentiment analysis.

4.2.1 Overview of the sentiment analysis

Table 1 Overview of the sentiment analysis done for other institutes in comparison to Lapworth

Institution	Sentiment Score	Positive Feedback	Negative Feedback
Lapworth Museum	0.28	Commended for educational value and specialized exhibits, in fossils and geology.	Issues raised about the museum's size, parking, and layout, which detract from the overall experience.
Manchester Museum	0.28	Recent renovations are acknowledged.	Visitors felt the changes were too elusive or did not meet expectations; certain exhibits perceived as lacking content.
Barber Institute of Fine Arts	0.26	Warm, welcoming atmosphere due to knowledgeable and friendly staff.	Criticism on the size of its collection.

4.2.2 Comparative Analysis: Lapworth Museum vs Manchester Museum

- **Visitor Experience:** The Manchester Museum, despite its larger scale, faces criticism for some exhibits being "empty" or "lacking content." This shows that although Manchester has the capacity to present a broad range of exhibitions, it occasionally falls short in curating these spaces effectively. A key takeaway for Lapworth is that, even with a smaller footprint, making sure each display is rich in information and attending to visitor comfort might greatly increase satisfaction.

Figure 7 Word cloud showing negative sentiment for Manchester Museum

- **Impact of Renovations:** Although there haven't been any recent significant changes at Lapworth Museum, the updates from Manchester Museum offer insightful

information. Manchester's renovations were meant to enhance the visiting experience, however numerous visitors failed to notice any significant changes after the restorations. This emphasizes how crucial it is to make upcoming improvements at Lapworth both practically sound and aesthetically pleasing, ensuring that visitors perceive a clear enhancement in their experience.

4.2.3 Comparative Analysis: Lapworth Museum vs. Barber Institute of Fine Arts

- **Specialization and Niche Appeal:** Lapworth Museum's focus on geology and fossils creates a strong niche appeal, attracting families and educational groups. This specialization is a significant strength, differentiating it from broader institutions like Manchester Museum. In a similar vein, the Barber Institute's emphasis on fine arts allows it to cultivate a dedicated audience. Both institutions succeed by offering deep, immersive experiences in their respective fields. Therefore, Lapworth's competitive edge depends on its aggressive maintenance and promotion of this specialty.
- **Visitor Interaction and Staff Engagement:** The Barber Institute excels in creating a warm, welcoming atmosphere, particularly through the friendliness and knowledge of their staff. This positive engagement greatly enhances the visitor experience. By benchmarking against the Barber Institute, Lapworth can enhance visitor engagement through more interactive sessions, guided tours, and better staff training, which would elevate overall satisfaction and help build a loyal visitor base.

Figure 8 Word cloud showing positive sentiment for Barber Institute

4.3 Visualization

Tableau effectively visualizes data, assisting with data-driven decision-making by delivering clear insights and supporting educated choices based on examined data (Hoelscher & Mortimer , 2018).

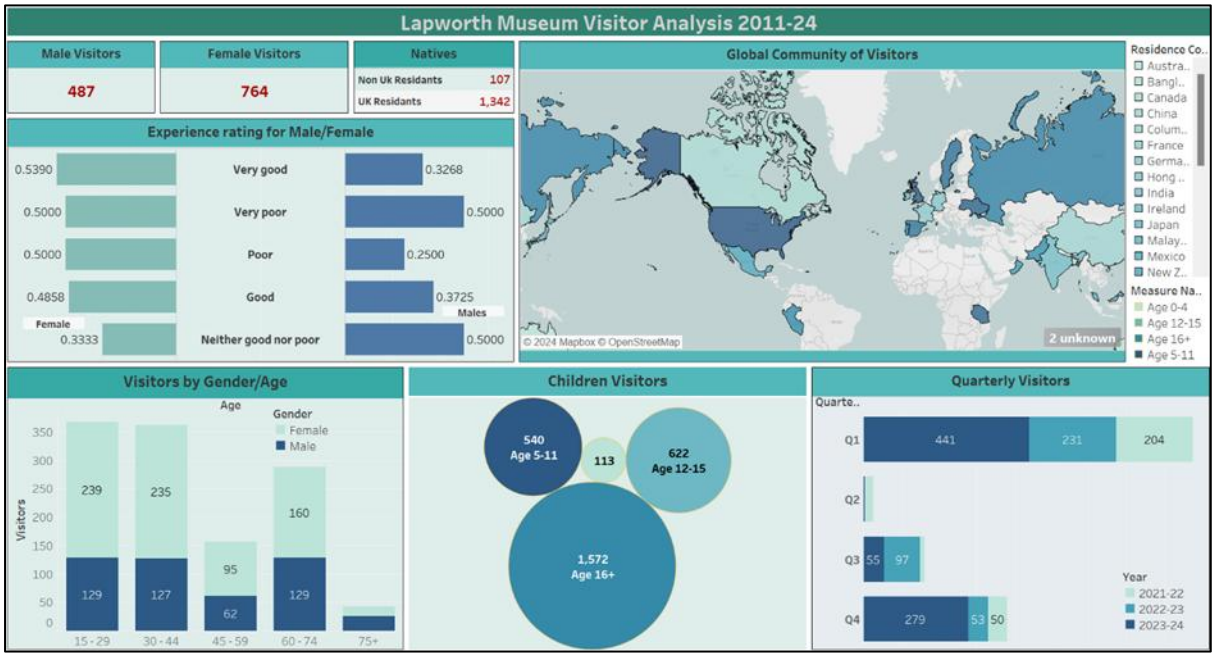

Figure 9 Dashboard showing Lapworth's demographics

[Capstone project 3 | Tableau Public]

- **Male Visitors & Female Visitors**
  The museum received 487 male visitor feedback, who made up a significant portion of the total visitor count. In contrast, 764 female visitors were recorded, indicating that the museum's exhibits and programs resonate more with female visitors. The museum could explore what drives higher female participation and seek to balance engagement across genders.
- **Natives (UK vs. Non-UK Residents)**
  The third group distinguishes between UK and non-UK inhabitants, with 1,342 UK visits against 107 overseas ones. This significant disparity suggests that most visitors are locals, which is understandable considering the museum's location. However, the presence of international visitors, although small emphasizes the museum's global reach and indicates potential to improve international marketing and design exhibitions that appeal to global audiences.

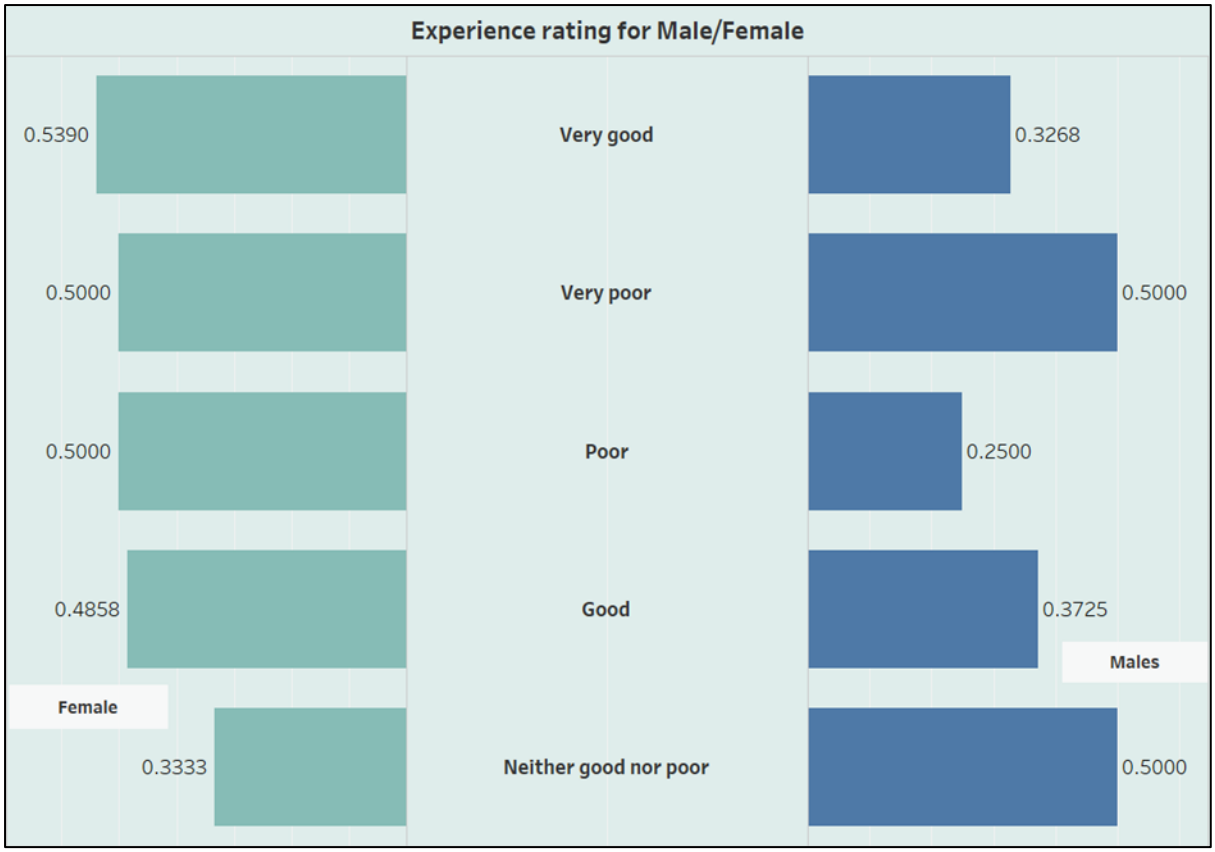

Figure 10 Butterfly chart showing experience rating by gender

Butterfly charts are very effective for evaluating two related datasets side by side. This enables simple visual comparisons of categories, such as demographic data between groups. (Bessler, 2023)**.**

The chart depicts differences in experience ratings between male and female visitors in several categories. It shows that a greater proportion of female visitors (53.9%) rated their experience as "Very Good," compared to 32.68% of male visitors.

In the "Good" category, reactions were roughly comparable, with 48.58% of females and 37.25% of males describing their experience as good. Notably, 50% of male visitors provided a neutral rating of "Neither Good Nor Poor," compared to 33.33% of females, implying that males had a more neutral experience.

On the bottom end of the scale, 50% of females assessed their experience as "Poor," compared to 25% of males, and both genders had an equal number (50%) of visitors rating their experience as "Very Poor."

This data suggests that female visitors had a more favourable experience, whilst male visitors were more likely to have a neutral or negative experience, providing insights that could help the museum target improvements and increase visitor satisfaction across genders.

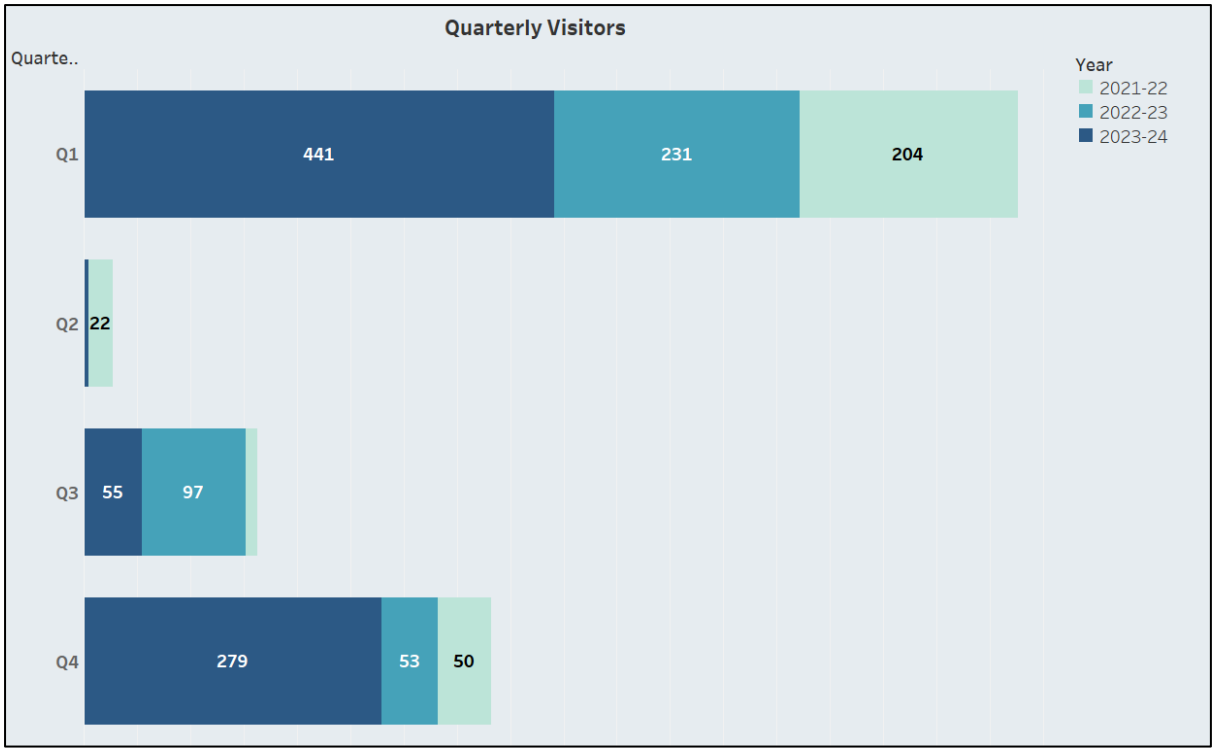

Figure 11 Number of visitors by quarters

The bar chart provides a comparative overview of visitor numbers across different quarters over three time periods: 2021-22, 2022-23, and 2023-24. The data shows that the first quarter consistently has highest visitor counts, with a substantial rise in 2023-24, reaching 441 visitors, up from 231 in 2022-23 and 204 in 2021-22.

This increasing trend suggests high early-year engagement. In contrast, the second quarter indicates a significant reduction in visitor numbers, notably in 2022-23, indicating a struggle in retaining interest. Q3 shows some improvement, with 97 visitors in 2022-23, but the number remains very low when compared to Q1.

Another increase is seen in Q4, particularly in 2023-24, with 279 visitors, indicating excellent end-of-year programming initiatives. Overall, the data shows high engagement in Q1 and Q4, with room for improvement in Q2 and Q3 to achieve more balanced attendance throughout the year.

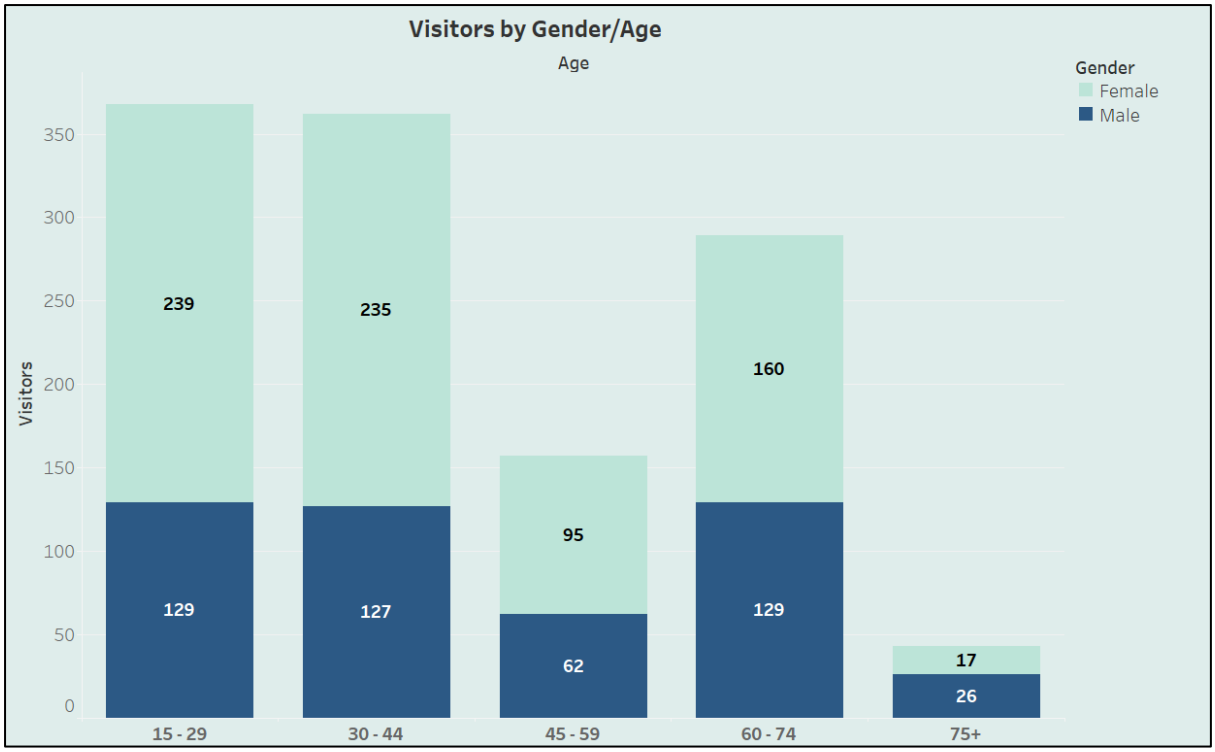

Figure 12 Bar chart depicting visitors by gender and age

Bar charts are emphasized as the major tool for displaying categorical data, which might include information from corporations, organizations, governments, or researchers (Bessler, 2023)**.**

The bar chart shows a breakdown of museum visits by age and gender. The most popular age groups are 15-29 and 30-44, with more female visitors (239 and 235, respectively) than men (129 and 127). This shows that the museum's displays are particularly appealing to young and middle-aged individuals, both male and female. The 60-74 age group also displays considerable engagement, demonstrating the museum's success in attracting older persons, most likely due to historical or cultural content.

In contrast, the 45-59 age bracket shows a considerable drop for both genders. This decrease may indicate a gap in the museum's programming for this cohort or reflect lifestyle issues such as work commitments that diminish their likelihood of visiting. Finally, the 75+ age group exhibits limited participation, with 17 female visits and 26 male visitors. This reduced turnout could be attributed to mobility challenges, or a lack of programs specifically geared toward this age group.

Overall, the figure shows that the museum's attraction is most among younger and middle-aged persons. However, there is still potential for improvement in terms of engaging visitors aged 45-59 and over 75, which might be addressed by personalized events or targeted outreach initiatives.

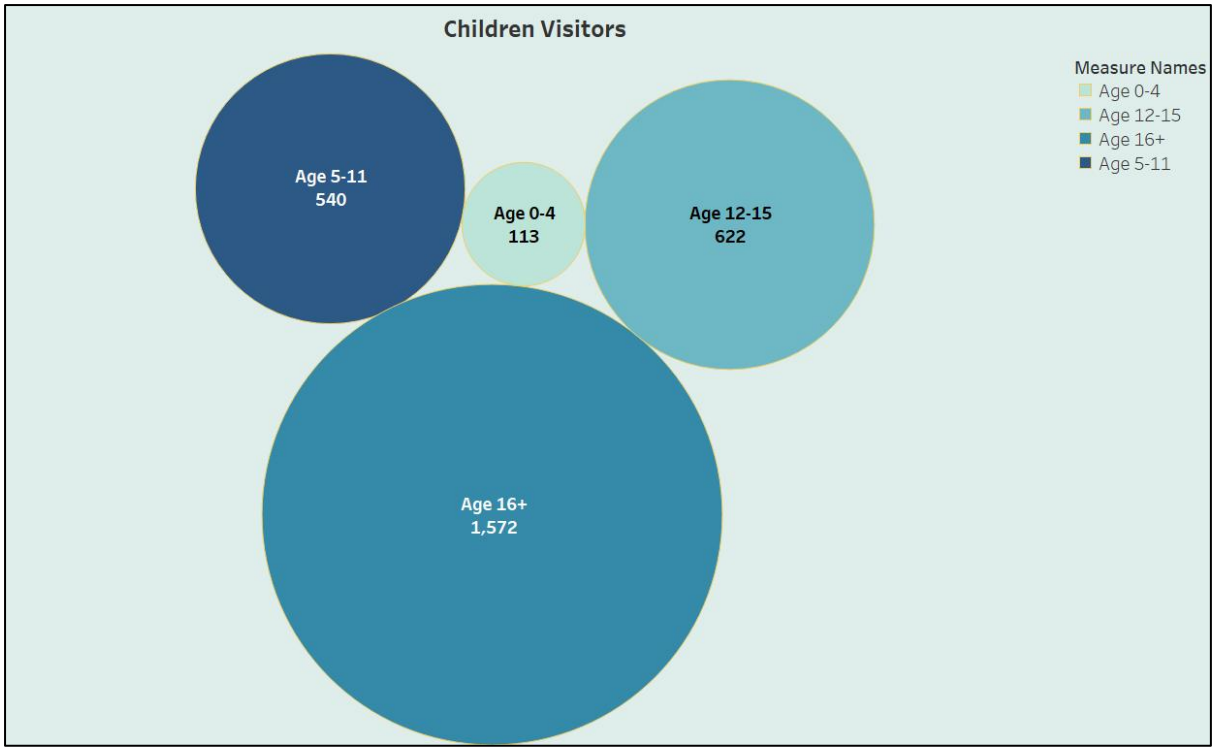

Figure 13 Bubble chart showing ages of children visitors

It can be observed that the majority of our young visitors are 16 and above. This shows that older teenagers and young adults are particularly interested in the exhibits and are perhaps drawn to more complex topics or interactive displays. The museum does, however, appeal to a varied spectrum of younger children, with 622 visitors in the 12-15 age group, 540 in the 5-11 age group, and 113 in the 0-4 age group.

Note: This data was provided anonymously by parents of the child to the museum. No data was directly collected from a child, keeping with the legal policy of the museum to not collect data from minors.

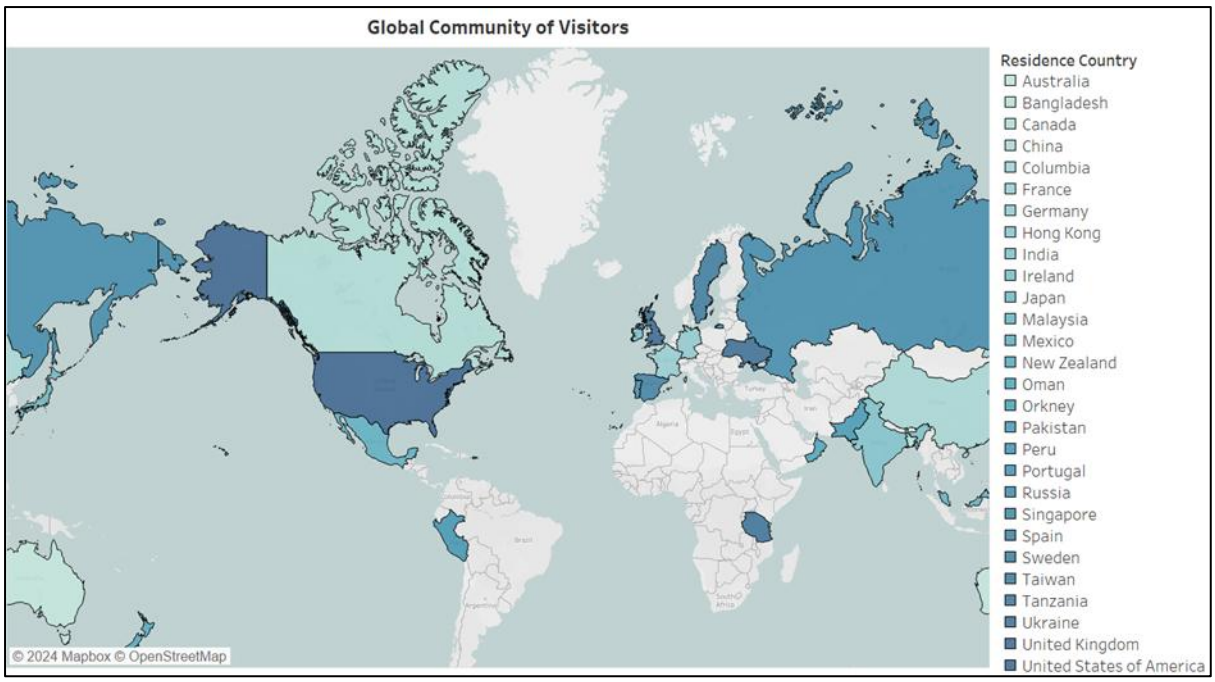

Figure 14 Map showing visitors from various countries

The map clearly depicts the museum's global reach, highlighting the diversity of its international audience. Each shaded country indicates a visitor's origin, demonstrating the museum's widespread geographic appeal. Geographical data representation uses GIS symbology, colour schemes, labelling, and visualizations to effectively convey information (Akowe & Ebite, 2023)**.** Colour schemes allow us to clearly distinguish one country from another.

The map highlights considerable representation from countries across multiple continents. The United States, Canada, and the United Kingdom are particularly well-represented, reflecting cultural ties, effective marketing, or the museum's strong reputation in these regions. Significant visitor numbers from France, Germany, Spain, Russia, and Ukraine demonstrate the museum's widespread appeal throughout Europe, including Eastern Europe. Representation from India, China, Japan, Malaysia, Singapore, and Taiwan highlights the museum's reach throughout Asia, especially Southeast Asia. Furthermore, visitors from Australia, New Zealand etc demonstrate the museum's global reach, which extends even to remote locations.

The diverse range of visitor origins demonstrates global interest in the museum's exhibits and programs. It recommends ways to increase international appeal through targeted outreach and personalized content. The data can uncover potential markets for expansion and deepen ties with already well-represented locations.

Maps did, in fact, result in faster problem-solving times, especially for more difficult problems, thanks to visual heuristics that simplify jobs and reduce errors as compared to tables (Smelcer & Carmel, 1997).

5. Recommendations

To enhance the visitor experience at the Lapworth Museum of Geology and ensure it continues to meet the evolving expectations of its diverse audience, we have developed a set of strategic recommendations. These suggestions are informed by a thorough analysis of visitor feedback, current museum practices, and a review of relevant literature. Our proposals are categorized into two groups: quick fixes, which can be implemented immediately to address immediate needs and improve operational efficiency, and long-term plans, which aim to elevate the museum's engagement and educational offerings. Each recommendation is designed to not only enhance the overall visitor experience but also to ensure the museum remains a dynamic, accessible, and educational environment for all guests.

5.1 Short-term Solutions

- **Improving Signage:** Museum signage plays a critical role in shaping the visitor experience. It encompasses directional signs that guide visitors through exhibits and informational signs that provide context to the displays. Effective signage is vital for enhancing visitor satisfaction, learning outcomes, and accessibility. (Karlita, 2011) in The Role and Effectiveness of the Signage System in Kirishima-Yaku National Park highlights that clear wayfinding systems reduce visitor anxiety, enabling a more focused engagement with exhibits.

  Their study found that visitors report a significantly more enjoyable experience when they can navigate the space without confusion. There is a strong correlation between the clarity and quality of signage and overall visitor satisfaction. Their findings suggest that improved signage significantly influences the likelihood of positive word-of-mouth recommendations and repeat visits.

- **Enhancing Parking Area:** Parking availability is a critical aspect of the logistical planning for museums and can significantly influence a visitor's decision to attend and their satisfaction with the overall experience.

  Convenient parking is significant in urban museums where alternative parking spaces may be costly or limited. A study by (Heng Zhang, 2018) in How Physical Environment Impacts Visitors' Behaviour in Learning-Based Tourism found that the first and last experiences of a museum visit—the ease of parking on arrival and departure—greatly affect overall visitor satisfaction. Their results indicate that visitor satisfaction drops significantly with the difficulty of finding parking.

- **Installation of Vending Machines:** Vending machines in museums are not just a source of refreshments; they play a strategic role in enhancing visitor convenience, extending the duration of museum visits, and providing a source of additional revenue.

  This review examines the multifaceted benefits of vending machines in the museum setting. (Wani, 2024) in the article Role of Vending Machines in Providing Public Facilities shows that the presence of vending machines can significantly enhance visitor satisfaction, especially among families with children and visitors who spend several hours in the museum.

  They argue that quick snack options allow visitors to manage their time more effectively within the museum. The findings suggest that vending machines represent a low-overhead

method of increasing ancillary revenues from visitors. The paper details that profit from vending machines can be reinvested into museum programs and services.

- **Offering Multilingual Support:** In an increasingly globalized world, museums attract visitors from diverse linguistic backgrounds. Offering multilingual support, such as guides, signs, audio tours, and information booklets in multiple languages, ensures that museums can effectively communicate with and engage a broader audience. (Ghazi, 2023) in the Journal of Museum Education states that Museums that invest in multilingual support demonstrate a commitment to accessibility, helping to bridge cultural and linguistic divides within their visitor base.

  The study found that non-native English speakers felt significantly more valued and welcomed when museums provided exhibit information in their native languages. It also examines how multilingual labels and interactive displays enhance the educational value of exhibits. The study shows that multilingual descriptions help convey cultural nuances more effectively to a diverse audience, enriching the learning experience.

- **Optimizing Online Presence:** In today's digital age, a museum's online presence is crucial for reaching and engaging with a global audience. Effective online strategies include developing interactive websites, utilizing social media platforms, and engaging in online marketing and digital storytelling. (Nuan Luo, 2022) in an article says that, Museums that actively engage in social media and online marketing are better positioned to enhance the visitor experience by fostering a sense of social presence and support, which in turn can lead to deeper visitor engagement.

  Their study shows that museums with interactive websites report higher engagement metrics and increased physical visitation rates. The impact of social media strategies on building online communities around museums. They find that consistent and interactive social media use helps museums develop loyal online followings and increases participation in museum programs and events.

5.2 Long-Term Plan:

- **Themed Exhibitions:** Themed exhibitions in museums are designed to offer visitors an organized and interpretive viewing experience, which often centres around a specific narrative, historical period, cultural phenomenon, or scientific concept.

  These exhibitions can enhance visitor engagement by making museum visits more relevant, accessible, and intellectually stimulating. (Leask, 2017) in the journal Museum Management and Curatorship say that Themed exhibitions effectively bridge the gap between education and entertainment, making museums more appealing to younger audiences, particularly Generation Y, by aligning with their preferences for social and interactive experiences. Themed exhibitions enhance the educational value of museum visits. By providing contextualized learning experiences, these exhibitions help visitors make meaningful connections between different exhibits and broader cultural or scientific themes.

- **Increasing Interactivity:** Interactive exhibits represent a dynamic approach within museums, designed to engage visitors actively in the learning process. These exhibits often utilize technology, hands-on activities, and immersive environments to facilitate deeper understanding and participation in the subject matter presented.

(Došen, 2021) state in the article that Interactive technology enhances visitor immersion by fostering both aesthetic and escapist experiences, making the museum visit more engaging and memorable also Museums that implement interactive exhibits see a significant increase in visitor satisfaction, as these technologies allow for deeper engagement with the content on display.

- **Establishing a feedback loop:** Feedback loops involve collecting, analysing, and acting on feedback from visitors to continuously refine and improve the museum experience. These mechanisms are essential for museums aiming to remain relevant and responsive to their audience's evolving preferences and expectations. (Falk, 2016) explores the impact of post-visit surveys on long-term visitor satisfaction.

  They find that museums that actively request and act on visitor feedback tend to have higher return rates and visitor loyalty. How feedback loops help museums optimize operational efficiency by identifying popular exhibits and potential bottlenecks within the museum space. This operational feedback allows for better resource allocation and crowd management.

5.2.1 Feedback loop

Another critical objective for this project was the feedback loop. The concept of a feedback loop is integral to continuous improvement in service-oriented industries, including museums.

According to (Viswanath, et al., 2019), an effective feedback loop should have these five key characteristics:

- **Agility and Flexibility:** It should adapt quickly to capture changes resulting from interventions.
- **Adaptiveness:** Interventions should be flexible and responsive to context-specific feedback.
- **Iterative Process:** It should involve multiple rounds of testing and feedback throughout implementation.
- **Responsiveness:** It must address the needs of implementers and decision-makers with a problem-driven approach.
- **Actionability:** The data collected should be relevant and timely for making informed design and implementation decisions.

Based on the characteristics of an effective feedback loop, we chose the Net Promoter Score (NPS) as the foundation for our feedback strategy for the museum. NPS aligns with the key traits required: it is flexible, allowing for survey questions to be tailored to the museum's needs, and it adapts to specific feedback contexts (Reichheld & Markey, 2011).

NPS works by asking visitors a single question to gauge their likelihood of recommending the museum, dividing them into Promoters, Passives, and Detractors. This classification helps museum staff pinpoint areas needing improvement. For instance, similar institutions have used NPS to enhance visitor experiences by focusing on specific feedback trends (Markey, et al., 2009).

Focusing on passives—those who are satisfied but not enthusiastic—can yield valuable insights. As noted by (Agag, et al., 2024), there's a significant relationship between a customer's promoter score (likelihood to recommend) and their online reviews or comments, also known as electronic word-of-mouth (eWOM).

While higher promoter scores generally result in more positive reviews, it's interesting to note that even those with scores of 9 (promoters) often share positive feedback, but individuals with scores of

7 (passives) provide a substantial amount of feedback as well, including both positive and negative comments. Despite this, companies tend to concentrate on detractors and promoters, frequently overlooking passives, who actually represent a large portion of overall feedback and can offer crucial perspectives on areas needing improvement.

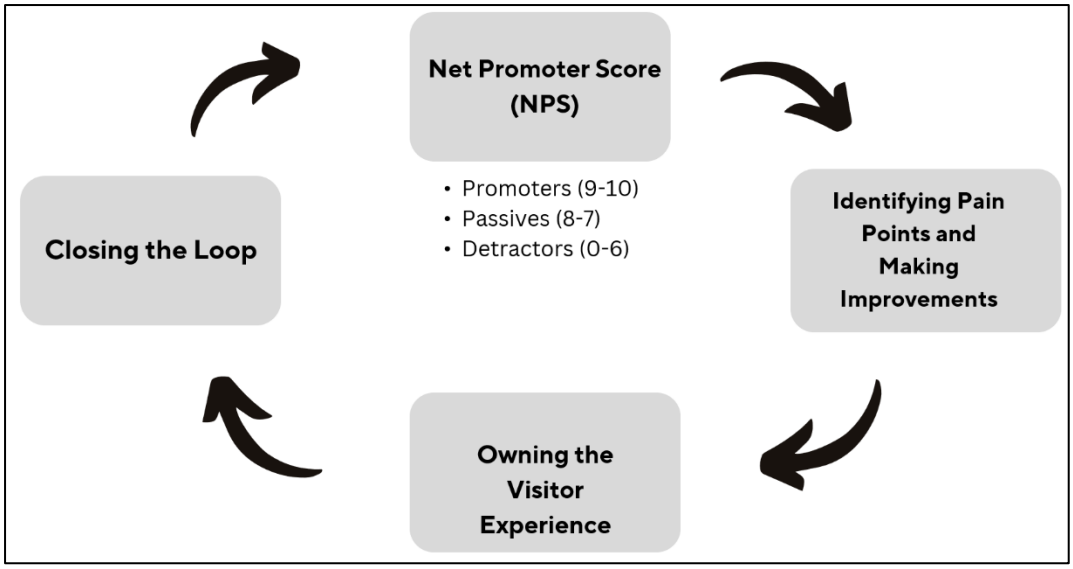

Figure 15 Feedback loop. Created by the author

The current paper-based survey system is costly and slow. We propose using Net Promoter Score (NPS) to measure visitor satisfaction at key points around the museum, such as specific sections or events. Visitors would be asked: "On a scale of 0 to 10, would you recommend the museum to friends and family?" Their responses categorize them into Promoters, Passives, or Detractors.

NPS is easy to implement and allows management to quickly identify areas for improvement. It offers a clear view of visitor sentiment and can highlight specific touchpoints affecting satisfaction, such as the time of day or particular exhibits. Focusing on Passives—those who are content but not enthusiastic—can provide valuable insights. Understanding why Passives rate the museum lower enables targeted improvements.

Implementing NPS can drive a shift toward a visitor-centred approach, prioritizing visitor needs over just the exhibits. Closing the feedback loop is essential: showing visitors how their feedback has led to tangible improvements builds trust and encourages ongoing engagement. Simple actions like website updates or social media shoutouts can demonstrate the impact of their input.

5.2.2 Building Community and Engagement within Visitors

Research highlights the importance of creating "moments of truth" or touchpoints with visitors, which can be as simple as a warm greeting upon entry. From our observations at the Barber Institute, we've learned that staff friendliness plays a crucial role in enhancing the overall visitor experience. These interactions, though brief, can either delight or alienate a visitor, making every touchpoint critical, especially when gathering feedback directly from the front line (Markey, et al., 2009).

Moreover, the significance of our online presence cannot be overstated. Electronic word-of-mouth (e-WOM) is a powerful tool; one study indicates that nearly 1 in 5 people decide against visiting a museum based solely on its website (Marty, 2007). A well-designed, informative website, along with an up-to-date Google Maps profile, can positively influence potential visitors even before they step

foot in the museum. Ensuring that our digital touchpoints are engaging and informative is essential to attracting and retaining visitors.

6. Conclusion

This project set out to evaluate the effectiveness of the Lapworth Museum's current strategies through sentiment analysis, benchmarking, and the implementation of a feedback loop. By analysing reviews from multiple platforms, including Google and TripAdvisor, we gained valuable insights into visitor perceptions and identified key areas for improvement. The sentiment analysis revealed that while the museum is generally well-received, there are specific aspects, such as its size and layout, that require attention to further enhance the visitor experience.

Benchmarking against similar institutions, such as the Manchester Museum and the Barber Institute of Fine Arts, demonstrated that despite limited resources, the Lapworth Museum performs on par with its peers. This finding underscores the museum's potential to maintain and even surpass its current standards with targeted improvements.

The integration of a feedback loop, grounded in the Net Promoter Score (NPS) system, offers a structured approach for the museum to continuously gather and act on visitor feedback. This iterative process not only supports ongoing enhancements but also fosters a culture of responsiveness and adaptability within the museum.

Overall, this project highlights the importance of using data-driven strategies to inform decision-making and drive continuous improvement in the museum sector. By adopting these approaches, the Lapworth Museum is well-positioned to enhance its offerings, increase visitor satisfaction, and ensure its long-term success.

7. References

Akowe, F. & Ebite, B., 2023. PAVEMENT MAINTENANCE MANAGEMENT USING GEOGRAPHIC INFORMATION SYSTEM (GIS) TECHNIQUES.

Bala Krishnamoorthy, C. D., January 2014. Benchmarking as a measure of competitiveness, s.l.: s.n.

Bessler, L., 2023. Bar Charts, Butterfly Charts, Waterfall Charts, Dot Plots, Needle Plots, Area Bar Charts, Text Graphs, and Line Charts: Charts for Categorical Data. Visual Data Insights Using SAS ODS Graphics, pp. 85-185.

Brinkworth, A., 2024. Lesser-known Birmingham attraction voted city's 'best hidden gem' and it's free, Birmingham: BirminghamLive.

Došen, T. K. a. Đ. O., 2021. Discovering the determinants of museum visitors' immersion into experience: the impact of interactivity, expectations, and skepticism. Current Issues in Tourism.

Falk, J., 2016. Museum audiences: A visitor-centered perspective. Society and Leisure.

Freytag, P. V. & Hollensen , S., 2001. The process of benchmarking, benchlearning and benchaction. The TQM Magazine, Volume 13, pp. 25-34.

Ghag, K. & Shah, K., 2015. Comparative analysis of effect of stopwords removal on sentiment classification. s.l., IEEE.

Ghazi, R., 2023. Translation Practices in Museums: What the Research Says. Journal of Museum Education.

Guzman, E. & Maalej, W., 2014. How Do Users Like This Feature? A Fine Grained Sentiment Analysis of App Reviews. s.l., IEEE, pp. 153-162.

Heimerl, F., Lohmann, S., Lange, S. & Ertl, T., 2014. Word Cloud Explorer: Text Analytics Based on Word Clouds. s.l., IEEE, pp. 1833-1842.

Heng Zhang, P.-C. C.-F. T., 2018. How Physical Environment Impacts Visitors Behavior in Learning-Based Tourism The Example of Technology Museum.

Hoelscher, J. & Mortimer , A., 2018. Using Tableau to Visualise data and drive decision-making. Journal of accounting education, Volume 44, pp. 49-59.

Jain, P. K., Pamula , R. & Srivastava, G., 2021. A systematic literature review on machine learning applications for consumer sentiment analysis using online reviews. Computer Science Review, Volume 41.

Karlita, Y., 2011. THE ROLE AND EFFECTIVENESS OF SIGNAGE SYSTEM. THE ROLE AND EFFECTIVENESS OF SIGNAGE SYSTEM IN KIRISHIMA-YAKU NATIONAL PARK.

Leask, P. B. a. A., 2017. Visitor engagement at museums: Generation Y and 'Lates' events at the National Museum of Scotland. Museum Management and Curatorship.

Liu, B., 2020. Introduction. In: Sentiment Analysis: Mining Opinions, Sentiments, and Emotions. Cambridge: Cambridge University Press (Studies in Natural Language Processing), pp. 1-17.

Markey, R., Reichheld, F. & Dullweber, A., 2009. Harvard Business Review. [Online].Available at: https://hbr.org/2009/12/closing-the-customer-feedback-loop. [Accessed 14 August 2024].

Marty, P. F. (2007) 'Museum Websites and Museum Visitors: Before and After the Museum Visit', Museum Management and Curatorship, 22(4), pp. 337–360. doi: 10.1080/09647770701757708.

McNaught, C., 2010. Using Wordle as a Supplementary Research Tool. The Qualitative Report, Volume 15.

Nuan Luo, Z. Z. Y. N. L. H., 2022. The impact of social media marketing activities in the museum industry.

Poria, S., Cambria, E., Hazarika, D. & Vij, P., 2016. A Deeper Look into Sarcastic Tweets Using Deep Convolutional Neural Networks. Proceedings of COLING.

Prasad, G. et al., 2024. Rule-Based Sentiment Analysis for Twitter(X) Data. s.l., IEEE.

Reichheld, F. & Markey, R., 2011. The Ultimate Question 2.0. Boston: Harvard Business Review Press.

Sandelowski, M., 2000. Focus on Research MethodsCombining Qualitativeand Quantitative Sampling,Data Collection,and Analysis Techniquesin Mixed-Method Studies. Research in Nursing & Health, Volume 23, p. 246–255.

Smelcer, J. B. & Carmel, E., 1997. The Effectiveness of Different Representations for Managerial Problem Solving: Comparing Tables and Maps. Decision Science Institute, 28(2), pp. 391-420.

Stevenson, W., 1996. Productions/Operations Management. 5 ed. s.l.:Irwin Publishing Company.

Ueland, B. a., 2016. Journal of Environmental Psychology.

Unwin, A., 2020. Why is Data Visualization Important? What is Important in Data Visualization?. Harvard Data Science Review, p. 1.

Viswanath, K., Synowiec, C. & Agha, S., 2019. Responsive feedback: Towards a new paradigm to enhance intervention effectiveness. Gates open research, Volume 3, p. 781.

Wani, M. A., 2024. Role of Vending Machines in Providing Public Facilities.

Zahoor, S. & Rohilla, R., 2020. Twitter Sentiment Analysis Using Lexical or Rule Based Approach: A Case Study. s.l., IEEE, pp. 537-542.

Zhang, Z., Wu, G. & Lan, M., 2015. ECNU: Multi-level Sentiment Analysis on Twitter Using Traditional Linguistic Features and Word Embedding Features. s.l., Association for Computational Linguistics, p. 561–567.